

Dr. Onur Çatmabacak

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PROFESSIONAL SUMMARY

Detail-oriented and results-driven **independent researcher** and **software developer** with 3 years of industry experience in product development. I have been an advanced **Linux** user and an enthusiast since 2008.

Proficient in BASH, PYTHON and FORTRAN programming languages. I have successfully developed and deployed commercial, scientific, and open-source software throughout my career.

I am experienced in **agile methodologies**, having worked in fast-paced environments where I consistently delivered projects on time and within budget. I am skilled at translating business requirements into technical specifications and possess strong problem-solving abilities to overcome complex development challenges. Furthermore, I am committed to writing clean, maintainable, and well-documented code, following best practices and industry standards. I continuously stay updated with emerging technologies and tools to enhance my development skills and stay at the forefront of innovation.

My articles have been cited $\sim$ **350** times and I have a **h-index of 9**.

EXPERIENCE

Linux System Engineer

Nov. 2023 - Present

Bank of Julius Baer

Zurich, CH

- Developing playbooks in Splunk SOAR platform for threat detection and response automation.
- Collaborate with a team of four to integrate data from various applications across Bank Julius Baer into Splunk, enhancing data management and analysis capabilities.
- Responsible for developing, maintaining, and enhancing dashboards, reports, and alerts within Splunk to improve data visibility and facilitate informed decision-making processes.
- Utilize the TrackMe add-on to monitor the health and stability of the bank's servers and infrastructure, ensuring optimal performance and reliability.
- Provide specialized technical support to various departments within the bank, demonstrating strong teamwork, a service-oriented approach, and effective communication skills across interdisciplinary teams.

Independent Researcher

Jan. 2023 - Present

Feza Gürsey Center for Physics and Mathematics

Istanbul, TR

- Physical modelling of quasi-periodic oscillations (QPOs) observed from the accretion disks of low-mass X-ray binaries.
- Spectroscopic modelling of active galactic nuclei (AGN) broad emission lines.
- Modelling of physical processes taking place in the accretion disks of ultra-luminous X-ray sources (ULXs).
- GPU parallelization of GIZMO astrophysical simulation code.

Software Engineer

Nov. 2021 – Jan. 2023

Spherene AG

Zurich, CH

- Replaced pymesh library with trimesh and many other various mesh libraries since the software license is not suitable for closed source applications and updated the python version of the whole application from 3.8.8 to 3.10.7. As a result, we did not pay +50k CHF for software licenses.
- Developed features, e.g. http1 to http2 upgrade and , for the back-end of our online service for additive manufacturing.
- Created a jupyter notebook utilizing the proprietary software developed by Spherene AG, running in a docker container on Google Cloud for an ESA project.
- Created projects running our closed-source additive manufacturing software on Google Cloud for the web service.
- Integrated FEniCS FEM Solver instead of NGSolve into our proprietary software.
- Replaced many packages with inconvenient software licenses for closed-source applications.
- Improved numerical algorithm of our closed-source additive manufacturing software.

PhD Researcher

Sep. 2016 – Dec. 2020

Institute for Computational Science, University of Zurich

Zurich, CH

- Developed a pipeline using Python to analyze large datasets of state-of-the-art cosmological simulations of the Universe from FIRE (Feedback In Realistic Environments) project that were run in various supercomputing centers (CSCS, XSEDE, BSC-CNS, Flatiron Institute, etc...) around the World.
- Proposed a theoretical model to explain the black hole - galaxy mass scaling relation.
- Created galaxy catalogs using Amiga Halo Finder (AHF), which is used by many other researchers around the world.
- Supervised our research group in IT, software development in Python/C/C++, and data analysis.

Supervised Projects

Institute for Computational Science, University of Zurich

Zurich, CH

- Artem Basyrov, Theoretical modeling of halo accretion, Master of Science Thesis Project, Feb. 2019 – May 2019
- Eric Rohr, The galaxy–halo size relation at cosmic noon, Think Swiss Summer Exchange Student Project, June 2019 – Sep. 2019

Teaching Assistant

Sep. 2016 – Feb. 2020

Institute for Computational Sciences, University of Zurich

Zurich, CH

- Teaching assistant for Theoretical Astrophysics, The Universe, Computational Astrophysics, Introduction to Data Science, Introduction to Computer Simulations courses.
- Responsible for problem-solving sections and correcting/grading weekly home-works.

Teaching Assistant

Sep. 2014 – June 2016

Faculty of Engineering and Natural Sciences, Sabanci University

Istanbul, TR

- Teaching assistant for Nature of Science course with ~ 750 students.
- Responsible for solving sessions and grading exams.

EDUCATION

University of Zurich

Zurich, CH

Ph.D. in Computational Astrophysics

Sep. 2016 – Oct. 2022

Thesis Title: The growth of supermassive black holes and their host galaxies in cosmological simulations.

Thesis Supervisor: Prof. Robert Feldmann

Kültür University

Istanbul, TR

M.Sc. in Physics, GPA:3.45

Sep. 2011 – June 2014

Thesis Title: Accretion disks around weakly magnetized neutron star low-mass x-ray binaries.

Thesis Supervisor: Assoc. Prof. M. Hakan Erkut

Kültür University

Istanbul, TR

B.Sc. in Physics with Honours, GPA:3.69

Sep. 2007 – June 2011

Thesis Title: Radial dependent one-dimensional analytical solution of viscous accretion disks around neutron star low-mass x-ray binaries.

Thesis Supervisor: Asst. Prof. M. Hakan Erkut

SKILLS

Languages: Turkish (Native), English (C1), German (B1)

Coding: Python ●●●●●, Django ●●●●●, BASH ●●●●●, FORTRAN95 ●●●●●, C/C++ ●●●●●, Julia ●●●●●

Codes: AHF ●●●●●, Gelato ●●●●●, BADASS3 ●●●●●, Fantasy AGN ●●●●●, PYQSOFIT ●●●●●, Sculptor ●●●●●, GIZMO ●●●●●

Dev Ops: Google Cloud ●●●●●, Docker ●●●●●, Git ●●●●●

Misc: Software License ●●●●● Linux ●●●●●

(**Grading:** Beginner ●●●●● Familiar ●●●●● Intermediate ●●●●● Comfortable ●●●●● Expert ●●●●●)

AWARDS AND SCHOLARSHIPS

- Second place in the "Sabanci University Academic Year Teaching Assistant Awards" 2016
- Full Merit Scholarship for Doctoral Studies, Sabanci University 2014 – 2016
- Full Merit Scholarship for Master Studies, Kültür University 2011 – 2014
- Full Merit Scholarship for Bachelor Studies, Kültür University 2007 – 2011
- Full Merit Scholarship for Bachelor Studies, Yeditepe University 2004 – 2007

WORKSHOPS & CONFERENCES

- Effective High-Performance Computing & Data Analytics with GPUs, Swiss National Supercomputer Centre (CSCS), Lugano/Switzerland, 15–25 July 2019
- Machine Learning for High Energy Physics - a mini course, , Physics Institute, University of Zurich, Zurich/Switzerland, 4-5 Feb. 2019.
- Scientific Programming with Python, Physics Institute, University of Zurich, Zurich/Switzerland, 25-29 June 2018.
- Directives Based GPU Programming, Swiss National Supercomputer Centre (CSCS), Lugano/Switzerland, 14-15 May 2018.
- AIM week, Academia Industrial Modelling week, ETH Zurich, Zurich/Switzerland, 7-11 Nov. 2016.

PUBLICATION LIST

16 papers (15 published, 1 in prep.), ~ 350 citations, h-index: 9

1. E. Nihal Ercan, E. A. Caliskan, M. Hakan Erkut, **Çatmabacak, O.**, Ö. Çatmabacak, A. Farhan, and E.P.J. van den Heuvel. A Possible Optical Counterpart of IeRASS J053448.4+212608. *in prep.*, July 2024
2. Robert Feldmann, Michael Boylan-Kolchin, James S. Bullock, **Çatmabacak, Onur**, Claude-André Faucher-Giguère, Christopher C. Hayward, Dušan Kereš, Alexandres Lazar, Lichen Liang, Jorge Moreno, Pascal A. Oesch, Eliot Quataert, Xuejian Shen, and Guochao Sun. Elevated UV luminosity density at Cosmic Dawn explained by non-evolving, weakly-mass dependent star formation efficiency. *arXiv e-prints*, page arXiv:2407.02674, July 2024
3. Lichen Liang, Robert Feldmann, Norman Murray, Desika Narayanan, Christopher C. Hayward, Daniel Anglés-Alcázar, Luigi Bassini, Alexander J. Richings, Claude-André Faucher-Giguère, Dongwoo T. Chung, Jennifer Y. H. Chan, Doğa Tolgay, **Çatmabacak, Onur**, Dušan Kereš, and Philip F. Hopkins. [C II] 158 μ m emission as an indicator of galaxy star formation rate. *MNRAS*, 528(1):499–541, February 2024
4. Robert Feldmann, Eliot Quataert, Claude-André Faucher-Giguère, Philip F. Hopkins, **Çatmabacak, Onur**, Dušan Kereš, Luigi Bassini, Mauro Bernardini, James S. Bullock, Elia Cenci, Jindra Gensior, Lichen Liang, Jorge Moreno, and Andrew Wetzel. FIREbox: simulating galaxies at high dynamic range in a cosmological volume. *MNRAS*, 522(3):3831–3860, July 2023
5. Jorge Moreno, Shany Danieli, James S. Bullock, Robert Feldmann, Philip F. Hopkins, **Çatmabacak, Onur**, Alexander Gurvich, Alexandres Lazar, Courtney Klein, Cameron B. Hummels, Zachary Hafen, Francisco J. Mercado, Sijie Yu, Fangzhou Jiang, Coral Wheeler, Andrew Wetzel, Daniel Anglés-Alcázar, Michael Boylan-Kolchin, Eliot Quataert, Claude-André Faucher-Giguère, and Dušan Kereš. Galaxies lacking dark matter produced by close encounters in a cosmological simulation. *Nature Astronomy*, 6:496–502, April 2022
6. **Çatmabacak, Onur**, Robert Feldmann, Daniel Anglés-Alcázar, Claude-André Faucher-Giguère, Philip F. Hopkins, and Dušan Kereš. Black hole-galaxy scaling relations in FIRE: the importance of black hole location and mergers. *MNRAS*, 511(1):506–535, March 2022

7. Eric Rohr, Robert Feldmann, James S. Bullock, **Çatmabacak, Onur**, Michael Boylan-Kolchin, Claude-André Faucher-Giguère, Dušan Kereš, Lichen Liang, Jorge Moreno, and Andrew Wetzel. The galaxy-halo size relation of low-mass galaxies in FIRE. *MNRAS*, 510(3):3967–3985, March 2022
8. Philipp Denzel, **Çatmabacak, Onur**, Jonathan Coles, Claude Cornen, Robert Feldmann, Ignacio Ferreras, Xanthe Gwyn Palmer, Rafael Küng, Dominik Leier, Prasenjit Saha, and Aprajita Verma. The lens SW05 J143454.4+522850: a fossil group at redshift 0.6? *MNRAS*, 506(2):1715–1722, September 2021
9. Alexandres Lazar, James S. Bullock, Michael Boylan-Kolchin, Robert Feldmann, **Çatmabacak, Onur**, and Leonidas Moustakas. Out of sight, out of mind? The impact of correlated clustering in substructure lensing. *MNRAS*, 502(4):6064–6079, April 2021
10. Lichen Liang, Robert Feldmann, Christopher C. Hayward, Desika Narayanan, **Çatmabacak, Onur**, Dušan Kereš, Claude-André Faucher-Giguère, and Philip F. Hopkins. The IRX- β relation of high-redshift galaxies. *MNRAS*, 502(3):3210–3241, April 2021
11. M. Hakan Erkut and **Çatmabacak, Onur**. Parallel Tracks as Quasi-steady States for the Magnetic Boundary Layers in Neutron-star Low-mass X-Ray Binaries. *ApJ*, 849(1):58, November 2017
12. M. Hakan Erkut, Şivan Duran, Önder Çatmabacak, and **Çatmabacak, Onur**. A New Correlation with Lower Kilohertz Quasi-periodic Oscillation Frequency in the Ensemble of Low-mass X-Ray Binaries. *ApJ*, 831(1):25, November 2016
13. M. Hakan Erkut, **Catmabacak, Onur**, Sivan Duran, and Önder Çatmabacak. Dependence of kHz quasi-periodic oscillation frequencies on accretion-related parameters. In *41st COSPAR Scientific Assembly*, volume 41, pages E1.1–27–16, July 2016
14. **Catmabacak, Onur** and M. Hakan Erkut. On the lack of correlation between X-ray flux and kHz quasi-periodic oscillation frequencies. In *41st COSPAR Scientific Assembly*, volume 41, pages E1.1–78–16, July 2016
15. Önder Çatmabacak, M. Hakan Erkut, **Catmabacak, Onur**, and Sivan Duran. Search for a correlation between kHz quasi-periodic oscillation frequencies and accretion-related parameters in the ensemble of neutron star low-mass X-ray binaries. In *41st COSPAR Scientific Assembly*, volume 41, pages E1.1–79–16, July 2016
16. P. Soffitta, R. Bellazzini, E. Bozzo, V. Burwitz, A. Castro-Tirado, E. Costa, and et al. XIPE: the x-ray imaging polarimetry explorer. In Jan-Willem A. den Herder, Tadayuki Takahashi, and Marshall Bautz, editors, *Space Telescopes and Instrumentation 2016: Ultraviolet to Gamma Ray*, volume 9905 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 990515, July 2016